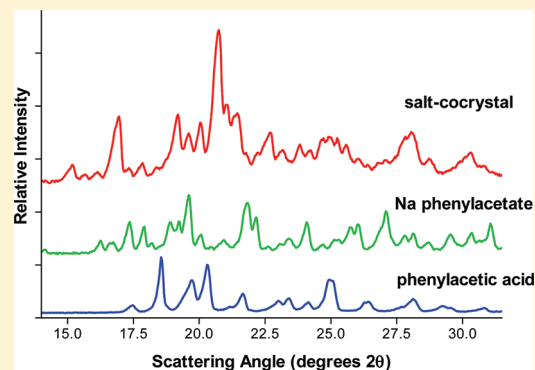


Cocrystal Systems of Pharmaceutical Interest: 2010

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ABSTRACT: The literature published during 2010 whose subject matter encompasses the cocrystallization of organic compounds having particular interest to pharmaceutical scientists has been summarized in an annual review. The papers cited in this review were drawn from the major physical, crystallographic, and pharmaceutical journals. After a brief introduction, the review is divided into sections that cover articles of general interest, the preparation of cocrystal systems and methodologies for their characterization, and more detailed discussion of cocrystal systems containing pharmaceutically relevant compounds. A brief summary of the state of the art of pharmaceutical cocrystals is also included, which poses an issue that is of great importance to the field.



1. INTRODUCTION

Pharmaceutical scientists are always interested in improving the physical properties of drug substances, and hence have delved into the scope of solid-state structural variations that can be obtained through the cocrystallization of several molecules in a single lattice structure.^{1–3} Along these lines, workers have researched the assembly of supramolecular synthons and crystal engineering in ever-increasing efforts to produce materials having new and useful properties.⁴ In the present review, the definition of a cocrystal proposed by Aakeröy will be used, namely, where cocrystal formation from supramolecular synthons is to be considered as forming from discrete neutral molecular species that are solids at ambient temperatures, and where the cocrystal is a structurally homogeneous crystalline material that contains the building blocks in definite stoichiometric amounts.¹

Actually, the cocrystal field has a long history, but historically mixed crystals were identified under other names, such as molecular complexes, association species, donor–acceptor complexes, and the like. Stahly has provided a history of cocrystals that were reported in the literature prior to 2000 and which contained only organic components, discussing their discovery and history, and illustrating principles of cocrystal chemistry through the use of illustrative examples.⁵ The present review continues the annual surveys corresponding to the literature published in 2007,⁶ 2008,⁶ and 2009,⁷ and will summarize literature published during 2010.

Although primary attention will be given to cocrystal systems having pharmaceutical interest, related papers that are of particular significance to the field will be discussed as well. The literature cited in the present review has been drawn from the major physical, crystallographic, and pharmaceutical journals, and consequently the coverage cannot be represented as being encyclopedic or comprehensive. Apologies are presented in advance to any scientist in the field whose works have been inadvertently omitted.

2. ARTICLES OF GENERAL INTEREST

A very useful review has been published that summarizes recent developments in the utility of cocrystals as means to improve the physical properties of pharmaceutical drug substances, including enhancement of drug solubility and dissolution rates, stability toward thermal and humidity stress, and material compressibility during tablet formation.⁸ Some of the other topics covered in this article include the design of supramolecular synthons, and the conduct of screening studies to evaluate stoichiometric variations in cocrystal composition. It was concluded that reliable strategies for the synthesis and design of cocrystal systems are effectively established, as well as the potential of cocrystallization for improvement of the solid-state properties of drug substances. The authors reported that workers in the field are seeking to develop a better understanding of cocrystal structure–property relationships through the conduct of systematic structural and computational studies.

Supramolecular synthons represent the fundamental building block of cocrystal systems, and the development in understanding of the scope and boundaries of the interactions responsible for formation of these synthons has been reviewed in detail.⁹ In this article, catemer and dimer $\text{HOC}=\text{O}\cdots\text{HOC}=\text{O}$ and $\text{C}=\text{O}\cdots\text{H}-\text{N}$ motifs were discussed, but more discussion was placed on interactions involving fluorine. Hence, the coverage of aggregation of organo-fluorine compounds, and supramolecular synthons based on $\text{C}-\text{F}\cdots\text{H}$ or $\text{C}-\text{F}\cdots\pi$ interactions, was extensive. Finally, the current state of crystal structure prediction was addressed in context with highly developed crystallization methodologies and the needed improvements in computational models.

The degree of proton transference in a hydrogen-bonded synthon determines whether a particular solid should be

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classified as a cocrystal or as a salt, and hence discussions of the continuum between salts and cocrystals retain their importance. The degree of transfer between substituted pyridine derivatives and a series of carboxylic acids has been evaluated using molecular electrostatic potential surface calculations, and it was reported that the calculated charges on the N-heterocyclic base-bond acceptor could be used to predict the existence of a salt or a cocrystal.¹⁰ Formation of the salt was inevitable once the charge on the hydrogen-bond acceptor exceeded a critical value, while the intended cocrystal could not form if the charge was too low.

In another study, Hammett substituent constants were used as the basis to model the formation of cocrystal products.¹¹ During the attempted cocrystallization of 32 acid/acid combinations of substituted benzoic acids, it was found that 90% of the systems formed cocrystals if the Hammett constants of the cofomers were of opposite sign, while only 25% of the systems where the constants were of the same sign yielded cocrystals. It was noted in the paper that a direct relationship exists between the Hammett constant and the ionization constant of the carboxylic acids, and hence systems characterized by large differences in substituent constants (and, by extension, large differences in pK_a) would be predicted to yield salts instead of cocrystals.

The formation of nicotinamide/salicylic acid and nicotinamide/oxalic acid cocrystal products was anticipated by principles of crystal engineering and trends in ionization constants, and the charge density distribution in these cocrystals has been studied.¹² In these structures, the C—H $\cdots$ O interactions were noted to provide support for the directional O—H $\cdots$ O and N—H $\cdots$ O hydrogen bonding. The two ionizations inherent to chloranilic acid were exploited to form both salt and cocrystal species with a variety of organic bases, characterized by six distinct motifs of hydrogen bonding.¹³ Since the structures contained protonated base functionalities, the dominant interaction was the N⁺—H $\cdots$ O hydrogen bond.

Another approach for studying the salt-cocrystal continuum has involved the evaluation of protonation states using X-ray photoelectron spectroscopy, where the shifting in energy of the nitrogen 1s spectrum of theophylline has been used to evaluate the degree of proton transfer. Conventional methods were used to demonstrate the formation of an adduct with citric acid, and the equivalence in energies of the nitrogen 1s spectra in theophylline and its citric acid adduct was taken as proof of the existence of a cocrystal and the lack of salt formation.¹⁴ This result is what one would have expected on the basis of the ΔpK_a rule. In a related study, the same methodology was used to demonstrate the formation of a theophylline cocrystal with oxalic acid, but only the formation of a salt species with 5-sulfosalicylic acid.¹⁵

One of the important rationales for modifying the properties of a drug substance is to improve its solubility and dissolution rate. Cocrystal eutectic constants, calculated as the ratio of the solution-phase concentrations at the eutectic point of the compounds making up a cocrystal species, have been found to serve as indicators of phase behavior.¹⁶ These constants were shown capable of providing guidance to the selection and synthesis of cocrystal systems. Hansen solubility parameters have been used as part of an interaction study of 30 cofomers with indomethacin, and their use was demonstrated to be a useful tool to improve the efficiency of cocrystal screening.¹⁷ The micellar solubilization of cocrystals has been found to follow a nonlinear dependence on surfactant concentration,

which was in turn related to the differential solubilization of the cocrystal components.¹⁸

In a double-blind study involving 10 systems, the ability to obtain accurate structures of cocrystals using high-resolution powder diffraction has been demonstrated.¹⁹ Even though conduct of the powder refinement methodology was significantly more difficult relative to the single-crystal method, the ability to obtain reliable structures of systems that do not yield adequate crystals is a great advantage in crystal engineering.

Dividing hydrates of cocrystal systems into four categories based upon thermal stability properties (i.e., water is lost at temperatures less than 100 °C, water is lost between 100 and 120 °C, water is lost at temperatures exceeding 120 °C, and where dehydration occurs concurrently with the melting transition), the implications for crystal engineering of water in cocrystals have been considered.²⁰ An analysis of structures in the Cambridge Structural Database revealed the existence of considerable diversity in the types of heterosynthons formed by water in the formation of hydrogen bonds with carboxylic acids or alcohols. It was concluded that the variability in supramolecular synthons involving water molecules is the factor that causes their unpredictable thermal stability.

3. PREPARATION OF COCRYSTAL SYSTEMS AND METHODOLOGIES FOR CHARACTERIZATION

Brittain has continued to develop vibrational spectroscopic selection rules for the recognition of salts and cocrystals, studying the sodium salt formation of benzoic acid, phenylacetic acid, hydrocinnamic acid, and 4-phenylbutanoic acid and in the 1:1 cocrystal products formed by the free acids and sodium salts.²¹ It was deduced from these studies that for salt and cocrystal systems involving carboxylic acid groups, the energy of the antisymmetric stretching mode of the carbonyl group could be used to identify the nature of the species formed. When the free acid absorption band (approximate frequency range of 1680–1690 cm^{-1}) disappeared entirely and became replaced by the corresponding anion band (approximate frequency range of 1550–1600 cm^{-1}), a salt has been formed. On the other hand, when the free acid absorption band undergoes a small shift toward higher energy (approximate frequency range of 1700–1730 cm^{-1}), one has observed the formation of a cocrystal. These trends are illustrated in Figure 1 for the phenylacetate system.

Several characterization techniques have been used to study the cocrystal products formed by salicylic acid with nicotinic acid, phenylalanine, and 6-hydroxynicotinic acid, as well as the cocrystal formed by 3,4-dihydroxybenzoic acid and oxalic acid.²² By completely assigning the Raman spectra of the cocrystals, a deeper understanding was obtained into the perturbation of vibrational mode frequencies upon formation of the supramolecular synthons. The use of in situ Raman spectroscopy has been reported, and shown to be an effective observation tool in the high-throughput screening of slurries containing indomethacin and 46 potential cocrystal formers.²³ It was reported that savings in cost and time could be used using the reported methodology, as the spectroscopic information could be rapidly obtained and analyzed during the course of the studies.

The utility of cocrystallization as a means to develop new routes for the crystallization of organic compounds has been demonstrated for the cinnamic acid–3-nitrobenzamide system.²⁴ The isolation of cinnamic acid from fermentation feedstocks was facilitated by formation of the cocrystal, as the reaction product formed at mole fractions that were seven times

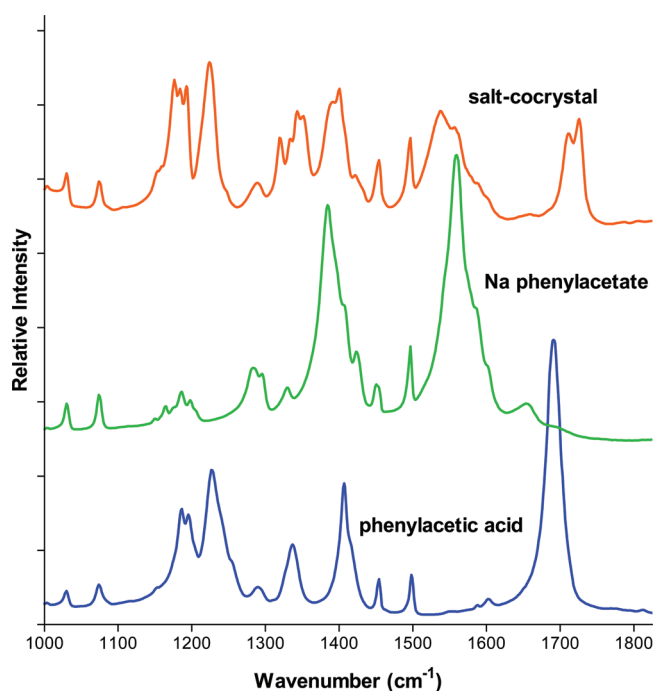


Figure 1. Infrared absorption spectra in the fingerprint region of phenylacetic acid (blue trace), sodium phenylacetate (green trace), and their 1:1 salt-cocrystal product (red trace) (adapted from ref 21).

less than the solubility of the pure acid. Using a simulated fed-batch system, it was shown that the cinnamic acid produced during the fermentation could be transformed into the less-soluble cocrystal. Eventually, a stationary state was obtained where the process could be operated near the 3-phase equilibrium point (i.e., between cocrystals, coformer solids, and dissolved solutes in solution).

The importance of phase diagrams to process technology has been demonstrated in two publications. Although little solution-phase interaction between benzoic acid and isonicotinamide was detected in 95% ethanol solutions, trends in the ternary phase diagram suggested differing crystal growth characteristics that were a function of composition.²⁵ In particular, excess benzoic acid appeared to increase the metastable zone width (yielding smaller crystals) and excess isonicotinamide decreased the metastable zone width (yielding larger crystals). A complete temperature-dependent ternary phase diagram was developed for the caffeine–glutamic acid–acetonitrile system, and subsequently used to develop a cooling cocrystallization process.²⁶ In this work, it was shown that cocrystal purity could be compromised when the crystallization took place outside of the defined safe operating region.

It has been shown that when one of the cocrystal formers has the ability to modulate the activity of water, modulation of the anhydrate–hydrate composition of the product can be obtained.²⁷ In particular, the effects of solution composition and water activity were studied for the theophylline/citric acid system, where it was reported that the critical water activity for the anhydrous–hydrate cocrystal equaled 0.8. In this work, it was also shown that the activities of both water and coformer determined the relative phase stability, and that excipients able to modify water activity could affect the eutectic points and phase stability. Finally, when cocrystals contain highly water-soluble cofomers, these products will be prone toward phase

transformations owing to the moisture uptake and deliquescence of the coformer.

The use of mixed solvent mixtures to suppress the formation of solvates during solution-phase cocrystallization has been demonstrated for the carbamazepine/saccharin system.²⁸ Using up to nine different solvate-forming solvents, and 18 known cocrystal formers of carbamazepine, it was reported that the probability of obtaining a nonsolvated product increased with the number of solvents used during the crystallization. The extreme method to obtain nonsolvated cocrystals is to generate the products either by a solvent-free reaction (or by a thermally induced combination of solid reactants) has been demonstrated for the systems formed by phenazine or acridine with vanillin.²⁹ Interestingly, while the phenazine/vanillin cocrystal was obtained by a solid-state reaction, the acridine/vanillin cocrystal could only be obtained from the melt.

While most cocrystal products are obtained on the small scale by solution-phase, mechanical, or solvent-drop methods, the use of alternate technologies continues to be investigated. An ultrasound-assisted solution-phase cocrystallization method has been studied for the noncongruently soluble system of caffeine and maleic acid.³⁰ Conventional methods yielded mixtures of caffeine and the 2:1 caffeine/maleic acid cocrystal in varying amounts, while only the ultrasound-assisted method yielded the pure 2:1 cocrystal product.

A supercritical fluid enhanced atomization process was used to produce cocrystals of various drug substances with saccharin.³¹ Cocrystal products with indomethacin, theophylline, caffeine, sulfamethazine, aspirin, and carbamazepine were produced from ethanol solutions using supercritical carbon dioxide as the atomization enhancing fluid, and the work served to demonstrate the utility of supercritical fluid technology as a means to obtain cocrystals of pharmaceutically important compounds. The authors also reported the formation of a 1:2 cocrystal of theophylline with saccharin that had not been previously reported.

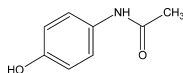
Cocrystals of six model systems were produced by spray drying, where the spray drying of solutions containing incongruently saturating solutes generated pure cocrystal products.³² It was suggested that the formation of cocrystals by spray drying represented a kinetically controlled route that minimized the effects of thermodynamics, namely, by proceeding through a noncrystalline form of the material. In another work, these same authors concluded that with sufficient understanding of the mechanism, the spray-drying process could be scaled up to a more useful level.³³

For cocrystals to achieve pharmaceutical significance, methods must be developed that enable the products to be manufactured on a commercial scale. One approach has been to use a solvent-free extrusion process, where enhanced surface contact between the cocrystal formers was achieved by a high-mixing screw design.³⁴ The utility of this method was communicated for the 2:1 caffeine/oxalic acid system, and for a 1:1 cocrystal of sorbic acid with an experimental compound. Spherically agglomerated 1:1 cocrystals of ibuprofen and nicotinamide were produced using a hot-melt extrusion procedure, where the effect of operating parameters (temperature, applied shear, and residence time) on the product properties was investigated using a number of techniques.³⁵ It was reported the process yield was improved by increases in mixing and shear intensity, with processing above eutectic point being required for cocrystallization to occur.

4. COCRYSTAL SYSTEMS HAVING PHARMACEUTICAL INTEREST

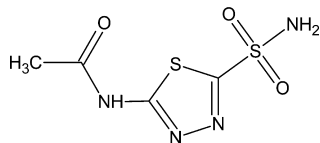
As cocrystal systems achieve more and more interest as potential solid-state forms of active pharmaceutical ingredients, the research conducted on such systems continues. In this section, the considerable amount of published work which has been conducted on cocrystal systems having direct interest to pharmaceutical scientists will be discussed.

An interesting cocrystal of **acetaminophen**:



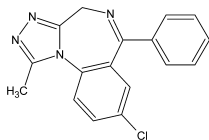
with 2,4-pyridinedicarboxylic acid has been reported, where although both coformer ingredients were white in color, the cocrystal product was red in color.³⁶ Analysis of the crystal structure revealed that the pyridinedicarboxylic acid converted to a zwitterionic form in the cocrystal, and that the cofomers assembled to form a three-dimensional hydrogen-bonded framework that featured a pronounced two-dimensional structure. Using density function calculations, it was concluded that the red color of the cocrystal was associated with a decrease in the bandgap separation between the highest-occupied and lowest-unoccupied molecular orbitals (i.e., the π - π^* separation) of the components in the supramolecular synthon relative to the analogous separation of the individual components.

Using solvent drop grinding procedures, 20 cocrystal formers have been combined with **acetazolamide**:



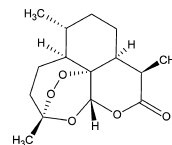
with these being identified by their X-ray powder diffraction patterns.³⁷ Single-crystal structural analysis was then used to study the hydrogen-bonding patterns in the products, and it was reported that while 4-hydroxybenzoic acid bound to the thiadiazole acetamide fragment of acetazolamide via C(N)-NH $\cdots$ HOOC and O-H $\cdots$ N interactions, nicotinamide bound via N-H $\cdots$ N and N-H $\cdots$ O interactions. Phase stability assays were conducted in water at physiological pH values, and it was reported that while some cocrystal products were stable, others underwent gradual transformations.

The heterosynthons present in the crystal structures of series of cocrystals and salts of **alprazolam**:



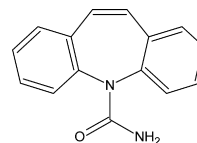
with carboxylic acids, boric acid, boronic acids, and phenols have been reported.³⁸ It was reported that the triazole ring acted as a hydrogen bond acceptor with the acidic cofomers, and the resulting structures contained a variety of hydrogen bonding patterns. Alprazolam was found to form hydrated structures when cocrystallized with aliphatic dicarboxylic acids where the waters of hydration played significant roles in the nature of the crystal structure. It was also reported that the triazole ring formed two different heterosynthons in its cocrystal with boric acid, while boronic acids formed non-centrosymmetric cyclic synthons and phenols formed O-H $\cdots$ N hydrogen bonds with the triazole ring.

While the relatively insoluble antimalarial drug **artemisinin**:



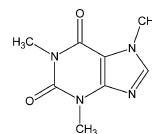
has found pharmaceutical utility, its lack of accessible ionizations has precluded salt formation as a means to enhance its solubility. However, using a mechanochemical screening procedure, cocrystal products of artemisinin have been formed with resorcinol (1,3-dihydroxybenzene) and orcinol (5-methylbenzene-1,3-diol).³⁹ Molecular descriptors derived from the Cambridge Structural Database were used to deduce whether cocrystals could be formed with artemisinin and 74 potential cofomers, and it was found that 33 of the cofomers would not be predicted to form a cocrystal product, thus enabling the use of a more focused screening process.

In spite of the large amount of work conducted on cocrystals of **carbamazepine**:



the compound continues to provide fertile ground for research. The structure of the isonicotinamide cocrystal Form-II was solved using powder diffraction data and found to be isostructural with the carbamazepine-nicotinamide cocrystal.⁴⁰ However, picolinamide did not form a cocrystal with carbamazepine, and so computed crystal energy landscapes and binary and ternary phase diagrams were used to explain this finding. It was concluded that the lattice energies of predicted carbamazepine-picolinamide cocrystal structures would be less than the lattice energies of the separated components, which was explained in terms of available intermolecular hydrogen-bonding capabilities.

Another cocrystal former that has received considerable interest is **caffeine**:

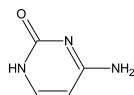


The energies of the known cocrystals of caffeine with 2- and 3-hydroxybenzoic acids were shown to be more stable relative to their separate components, as was the instability of any possible cocrystal between caffeine and the more stable conformer of 3-hydroxybenzoic acid. The ability of caffeine to form a stable cocrystal with 4-hydroxybenzoic acid was confirmed in this study, and the same cocrystal system was studied in greater detail in a separate study conducted by different authors.⁴¹ In this latter work, the use of differing initial concentration ratios yielded cocrystals of differing stoichiometry, suggesting that the microscopic intermolecular interactions determined in the solution state could serve as qualitative and predictive indicators for the final crystalline products. In another study, sonochemistry and a surfactant were used to produce nanosized cocrystal products of caffeine with 2,4-dihydroxybenzoic acid.⁴²

A 1:1 cocrystal of caffeine with 2-hydroxy-1-naphthoic acid was isolated upon execution of a screening procedure that was based on solution-mediated phase transformation.⁴³ The components were connected by means of hydrogen bonding that involved both an intermolecular O-H $\cdots$ N hydrogen bond

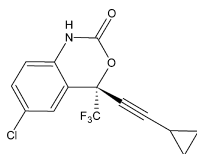
and an intramolecular O–H...O hydrogen bond. In another study, cocrystals with maleic acid having 1:1 and 2:1 stoichiometry have been obtained, and characterized through measurement of the binary and ternary phase diagrams.⁴⁴ The differing cocrystal products formed by caffeine with mono-hydroxybenzoic acids have been studied as to their computed crystal energy landscapes to rationalize the diversity.⁴⁵

The structures and properties have been reported for three cocrystals formed by dicarboxylic acids with **cytosine**:



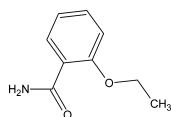
have been reported.⁴⁶ Solvent-drop grinding and solution-phase crystallization procedures were used to obtain the 4:1 cocrystal of cytosine with oxalic acid, the 2:1 cocrystal of cytosine with malonic acid, and the 2:1 cocrystal of cytosine with succinic acid. The same $R_2^2(12)$ structural motif was detected in the three products, which was formed by a pair of N–H...O hydrogen bonds within a dimer of cytosine.

Various solid-state conformations of **efavirenz**:



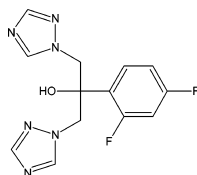
were found in its pure crystal form, its cyclohexane solvatomorph, and in cocrystals with 1,4-cyclohexanedione and 4,4'-bipyridine.⁴⁷ The formation of cocrystal products by this drug substance proved to be unpredictable, as only the two cocrystals were obtained after performance of at least 27 screening experiments. Two distinct heterosynthons were identified in the structure of the 4,4'-bipyridine cocrystal: one of the efavirenz molecules interacts with the bipyridine through cyclic N–H...N/C–H...O hydrogen bonds, and the other efavirenz molecule is joined to the other end of the bipyridine molecule through a single-point N–H...N hydrogen bond.

Two enantiotropically related polymorphic forms of the 1:1 cocrystal formed by ethylmalonic acid with **ethenzamide**:



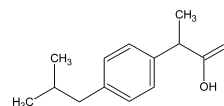
have been reported and characterized by a variety of analytical techniques.⁴⁸ The single-crystal X-ray diffraction study revealed that in both polymorphs the ethylmalonic acid coformer molecules adopted different molecular conformations even though the same hydrogen-bonding motif was present in both crystal types. In another study, the 1:1 cocrystal of ethenzamide with 3,5-dinitrobenzoic acid has been shown to exist in two nonsolvated polymorphs and in a series of solvates.⁴⁹ In a statistical study with structures filed in the Cambridge Structural Database, it was deduced that the tendency to form solvatomorphs was significantly higher for cocrystals as compared to the crystalline solvatomorphs of a single solid component.

The carboxylate OH...N_{arom} heterosynthon of **fluconazole**:



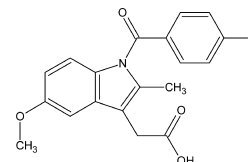
was exploited in a cocrystal design strategy in order to improve the physicochemical properties of this antifungal agent.⁵⁰ Cocrystal products were obtained of fluconazole with maleic, fumaric, and glutaric acids upon solvent evaporation, and the results of single-crystal structural analysis were used to understand the intermolecular connectivities in the respective supramolecular synthons. In the maleic acid product, evidence for proton transfer from the acid to fluconazole was obtained even through the pK difference between the coformers was less than that expected for salt formation.

The first reported example of cocrystal solid solutions for a system containing enantiomers of opposite chirality has been reported for the 1:2 product formed by 4,4'-dipyridyl and **ibuprofen**.⁵¹



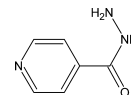
The enantiomeric mixtures of (ibuprofen)₂(4,4'-dipyridyl) cocrystals were obtained as a Roozeboom type-II solid solution, with the crystal structures of the solids being similar to those of their corresponding enantiomorphous and racemic coformers. It was found that the cocrystal products exhibited significantly higher melting temperatures and enthalpies of fusion relative to their parent ibuprofen chiral crystals, and the solid solutions could be crystallized to obtain a chiral enrichment. This latter feature suggested that cocrystallization could be used to obtain alternate methods for the chiral resolution of racemic compounds.

Since the cocrystal of saccharin with **indomethacin**:



has been shown to exhibit more solubility than indomethacin alone at all pH values, a study was conducted to evaluate the in vitro dissolution and in vivo bioavailability of the cocrystal product relative to a simple physical mixture and a commercial product.⁵² It was reported that the dissolution rate associated with the cocrystals was higher relative to that of other simpler forms of indomethacin, and that the in vivo bioavailability in dogs was also higher for the indomethacin–saccharin cocrystal. This study is one of the few that has provided a case study where the properties of a poorly soluble drug substance were improved upon formation of a pharmaceutically acceptable cocrystal.

The structures of a series of cocrystals of the antituberculosis drug **isoniazid**:

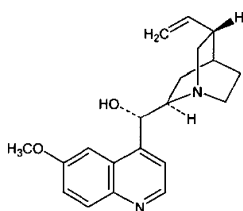


with 4-hydroxybenzoic acid, 2,4-dihydroxybenzoic acid, malonic acid, succinic acid, glutaric acid, adipic acid, and pimelic acid have been reported.⁵³ A search of literature cited in the Cambridge Structural Database revealed the scope of possible homosynthons and heterosynthons that would be most plausible in cocrystals of isoniazid with mono- and dicarboxylic acids. In the cocrystal systems of isoniazid, the dominant interaction was the COO...HN hydrogen bond, which was identified in the structures of all seven cocrystals of this study.

Nc1nc(N)nc(C2=CC=CC=C2Cl)nc1CC1=CC=C(C=C1)C(=O)NCCNCCFC(F)(F)c1nc2cc(C(F)(F)F)ccc2c1C(O)C3CCCCCN3
HClCc1cc(NC(=O)c2c(O)c3ccccc3s2(=O)N2C=C2)nn1Nc1nc(N)nc(N2CCCCC2)n1COc1ccc(cc1C=Cc2ccc(Cl)cc2)OCCOC(=O)c1c(C)c2c(c1)c(c3ccccc23)C(=O)N(C)CNC(=O)CN1CCCC1=O

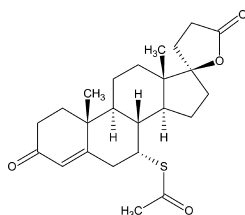
products were obtained by crystallization from acetonitrile. Interestingly, a cocrystal could not be obtained with 2,6-dihydroxybenzoic acid, which was ascribed to steric effects of the hydroxyl groups and preferred intramolecular hydrogen bonding between the hydroxyl group and the carboxylic acid group. Structural and infrared spectroscopic analyses of the five cocrystal products indicated that the patterns of hydrogen bonding in the various products were quite different.

The applicability of the O–H...N heterosynthon for synthesis of cocrystals was examined for the case of methyl paraben and **quinidine**:



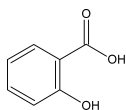
In this work, the authors introduced the concept of a “molecular hook” to describe how methyl paraben picked its target site of interaction via hydrogen-bond-mediated molecular recognition.⁶² The local conformation and hydrogen-bonding patterns were derived from advanced multinuclear solid-state nuclear magnetic resonance techniques, where the interpretation of the NMR data was supported by density-function quantum-chemical calculations. One interesting finding of the study was that the molecular specificity of methyl paraben enabled the stereoselective separation of quinidine from its quinine stereoisomer.

The hemihydrate, and the dehydrated form, of the 1:1 cocrystal formed by saccharin and **spironolactone**:



have been studied, as well as the in vitro dissolution of the hemihydrate.⁶³ Structural analysis indicated that the hemihydrate water resides in linear channels, and also that the hemihydrate and its anhydrate were isostructural. Single-crystal X-ray diffraction using the cocrystal anhydrate prepared from single-crystal to single-crystal dehydration confirmed the isostructural relationship, and showed the retention of the channels and shrinkage of the unit cell upon dehydration. While the hydrated cocrystal demonstrated improved solubility relative to spironolactone Form-II (the most stable form), over time the solubility decreased upon formation of the previously unreported spironolactone 0.33-hydrate.

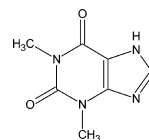
The cocrystal formed by benzamide with **salicylic acid**:



has been studied by a series of physical measurements, and it was determined that Raman spectroscopy was particularly useful for the characterization of the products and for a study of the nature of the interactions in the cocrystallized product.⁶⁴ Changes in the intensities of the amide and the carboxylic acid vibrational modes were noted to accompany formation of the

cocrystal, and new vibrational bands were identified in the cocrystal product that could be used as identifiers of cocrystal formation.

The 1:1 cocrystal formed by methyl gallate and **theophylline**:



has been studied, as well as the tableting performance of the cocrystal and its coformers.⁶⁵ While it was found that cocrystallization improved the tableability of methyl gallate, it was found to significantly deteriorate the tableability of theophylline, a difference ascribed to the dissimilar degrees of crystal plasticity and elasticity. In another work, the 1:2 cocrystal formed by theophylline and sulfamethazine was obtained using dry cogrinding, solvent-drop grinding, and slow evaporation procedures.⁶⁶ The structure was obtained using single crystal X-ray diffraction data, showing that the sulfamethazine molecules form a dimer through intermolecular hydrogen bonding (O...H–N) and that two intermolecular hydrogen bonds (O...H–N and N...H–N) attach theophylline to the dimer. The product formed by cocrystallization of with 5-sulfosalicylic acid studied by X-ray photoelectron spectroscopy (XPS) was identified as a salt, with the XPS method being used to determine the protonation state of the nitrogen functional groups on the theophylline moiety.⁶⁷

5. PHARMACEUTICAL COCRYSTALS: STATE OF THE ART

Over the past few years, work in the pharmaceutical community has begun to change the image of cocrystals from that of crystallographic curiosities to viable and useful forms of drug substances. Given that many new drug substances under development exhibit solubility limitations, workers have begun to successfully use cocrystallization as a means to enhance solubility and dissolution rate. Others have noted that the use of appropriate cocrystal forms can enhance the stability of a drug substance in the solid state and have exploited this feature to improve the long-term quality of an active pharmaceutical ingredient. Areas yet to be explored concern how cocrystallized drug substances will facilitate the formulation of such materials into viable dosage form, but there is no doubt that such work will follow.

As of yet, the regulatory hurdle toward the use of cocrystals in pharmaceutical products remains to be resolved. While agencies view polymorphs and solvatomorphs as alternate solid-state forms of drug substance and routinely accept their use in dosage forms, new issues are generated by the fact that cocrystallization necessarily results in a material in which the active pharmaceutical ingredient and another substance are in intimate contact at the molecular level. Therefore, the question to be addressed is whether a cocrystal should be defined as a physical mixture (and thus possibly being able to be classified within current compendial guidelines), or whether a cocrystal is actually a new chemical entity that requires as much safety and toxicology testing as would any other new chemical entity. It is fairly certain that investigators seeking to use cocrystals in pharmaceutical dosage forms will have to face this situation at some point during a regulatory review process, and this possibility may require some degree of intestinal fortitude and courage on the part of the company filing the drug application.

Nevertheless, if the benefits of cocrystallization are sufficient, one should be able to overcome any additional regulatory issues.

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